Natural Language Processing

100 Questions: Pre-processing, BoW, TF-IDF, Naïve Bayes, Evaluation

GARP RAI Exam Preparation

Q1: NLP Definition - Tutorial

QUESTION: What is Natural Language Processing (NLP)?

OPTIONS:

1. A technique for processing numerical data
2. An aspect of machine learning concerned with understanding and analyzing human language, both written and spoken
3. A method for image recognition
4. A database management technique

ANSWER: B

DETAILED TUTORIAL:

- **Definition:** NLP is the intersection of computer science, artificial intelligence, and linguistics
- **Core purpose:** Enable computers to understand, interpret, and generate human language
- **Key challenge:** Most information exists in unstructured, free text format
- **Alternative names:** Content analysis, text mining, computational linguistics
- **Why it's important:**
  - Unstructured text is the most abundant form of data
  - Manual analysis is slow, expensive, and inconsistent
  - NLP enables automated analysis at scale

EXAM TRAP: NLP is about understanding human language - not just processing text!

Q2: NLP vs Numerical Data - Tutorial

QUESTION: Why is NLP more challenging than modeling purely numerical data?

OPTIONS:

1. Numerical data is always larger
2. Textual data is often noisy with errors and ambiguities
3. NLP requires more computing power
4. Numerical data is more interesting

ANSWER: B

DETAILED TUTORIAL:

- **Key challenges in NLP:**
  1. **Ambiguity:** Same words have different meanings (polysemy)
  2. **Context dependence:** Meaning depends on surrounding text
  3. **Noise:** Spelling errors, abbreviations, colloquialisms
  4. **Informal language:** Social media, text messages
  5. **Sarcasm and tone:** Hard for machines to detect
- **Examples:**
  - "u r gr8" vs "you are great"
  - "lol" can mean different things in different contexts
  - "That's just great!" - could be sincere or sarcastic
- **Impact:**
  - NLP may fail to fully capture document essence
  - Power and flexibility are improving rapidly
  - Advances in computational capacity help

EXAM TRAP: NLP challenges include ambiguity, noise, and context - not just technical issues!

Q3: NLP Applications in Finance - Tutorial

QUESTION: Which of the following is an NLP application in finance?

OPTIONS:

1. Fraud detection using SEC filings
2. Customer service call routing
3. Sentiment analysis from social media
4. All of the above

ANSWER: D

DETAILED TUTORIAL:

- **Key NLP applications in finance:**
  1. **Fraud detection:**
     - SEC used NLP to detect accounting fraud
     - Analyze financial reports for inconsistencies
  2. **Call routing:**
     - Keyword recognition for customer inquiries
     - Automated triage to appropriate departments
  3. **Sentiment analysis:**
     - Social media monitoring
     - Market reaction assessment
     - Academic studies on asset prices and sentiment
  4. **Readability assessment:**
     - Fog index measurement
     - Plain English compliance
  5. **Document similarity:**
     - Assessing new information content
     - Tracking disclosure changes

EXAM TRAP: NLP has many financial applications - fraud detection, sentiment, customer service!

Q4: NLP vs Manual Reading - Tutorial

QUESTION: What are the main benefits of NLP over manual document reading?

OPTIONS:

1. NLP is always more accurate than humans
2. Superior speed, no scope to miss aspects, and consistent application
3. NLP is cheaper than human labor
4. NLP doesn't make mistakes

ANSWER: B

DETAILED TUTORIAL:

- **Key advantages of NLP:**
  1. **Speed:**
     - Processes thousands of documents quickly
     - Far faster than human reading
     - Enables real-time analysis
  2. **Completeness:**
     - No scope to miss any aspects
     - Provided they are built into the design
     - Consistent across all documents
  3. **Consistency:**
     - All documents considered identically
     - No scope for biases
     - No inconsistencies in application
- **Limitations:**
  - NLP may miss nuance
  - Ambiguity remains challenging
  - Requires careful design

EXAM TRAP: NLP advantages = speed, completeness, and consistency!

Q5: NLP Data Sources - Tutorial

QUESTION: What are the most common NLP data sources in finance?

OPTIONS:

1. Only financial statements
2. Corporate disclosures, social media posts, and news wires
3. Only social media
4. Only news articles

ANSWER: B

DETAILED TUTORIAL:

- **Primary data sources:**
  1. **Corporate disclosures:**
     - Mandatory filings (10-Q, 10-K)
     - Equity and bond issue prospectuses
     - Formal, structured language
  2. **Social media posts:**
     - Generalist: X (Twitter), Reddit
     - Focused financial: Stocktwits, Seeking Alpha
     - Informal, noisy, colloquial
  3. **News wires:**
     - Wall Street Journal
     - Dow Jones newswire
     - Formal, professional language
- **Data collection methods:**
  - API (Application Programming Interface)
  - Web scraping with Python
  - Batch downloads of PDFs
- **Corpus:** Collection of all retrieved documents

EXAM TRAP: Common sources = corporate disclosures, social media, and news wires!

Q6: JPMorgan COiN Platform - Tutorial

QUESTION: What did JPMorgan Chase's COiN platform achieve?

OPTIONS:

1. It made no significant impact
2. Significantly reduced time required to review legal documents
3. It replaced all human reviewers
4. It only worked for English documents

ANSWER: B

DETAILED TUTORIAL:

- **COiN (Contract Intelligence) Platform:**
  - Automated review of legal documents
  - Focus on commercial loan agreements
- **Challenges addressed:**
  - Manual review was time-consuming
  - Manual review was prone to errors
  - Extracting information from unstructured text
- **Outcomes:**
  - Significantly reduced review time
  - Improved consistency in information extraction
  - Improved accuracy in information extraction
- **Training:**
  - NLP models trained on annotated legal documents
  - Performance improved over time

EXAM TRAP: COiN reduced legal document review time significantly!

Q7: Other NLP Use Cases - Tutorial

QUESTION: Which institutions use NLP for fraud detection?

OPTIONS:

1. Only JPMorgan Chase
2. HSBC uses NLP for fraud detection
3. Only the SEC uses NLP
4. No financial institutions use NLP for fraud detection

ANSWER: B

DETAILED TUTORIAL:

- **NLP use cases in banking:**
  1. **Fraud detection:**
     - HSBC uses NLP for fraud detection
     - Analyze patterns in financial transactions
  2. **Customer service:**
     - Bank of America (BoA) chatbots
     - Automated customer interaction
  3. **Market sentiment:**
     - BlackRock uses NLP for sentiment analysis
     - Analyze market sentiment from news and social media
- **Early examples:**
  - SEC used NLP to detect accounting fraud
  - Pioneered the use of NLP in finance

EXAM TRAP: HSBC = fraud detection, BoA = chatbots, BlackRock = sentiment analysis!

Q8: Corpus Definition - Tutorial

QUESTION: What is a corpus in the context of NLP?

OPTIONS:

1. A single document
2. The collection of all documents retrieved for analysis
3. A type of machine learning algorithm
4. A data pre-processing technique

ANSWER: B

DETAILED TUTORIAL:

- **Definition:**
  - From Latin "corpus" meaning "body"
  - Collection of all documents available for analysis
  - May include hundreds or thousands of documents
- **Characteristics:**
  - All documents in the corpus are processed together
  - Vocabulary built from the corpus
  - Analysis results apply to the corpus
- **Examples:**
  - All 10-K filings of S&P 500 companies
  - All tweets about a specific stock
  - All customer service emails

EXAM TRAP: Corpus = collection of all documents - not a single document!

Q9: NLP Pre-processing Importance - Tutorial

QUESTION: Why is data pre-processing important in NLP?

OPTIONS:

1. It's not important
2. To convert raw text into a format suitable for analysis
3. To make documents shorter
4. To remove all words

ANSWER: B

DETAILED TUTORIAL:

- **Purpose of pre-processing:**
  1. Convert text to machine-readable format
  2. Remove non-textual information (HTML, tags)
  3. Standardize the text
  4. Reduce noise and improve accuracy
- **Initial cleaning steps:**
  - Extract text from PDFs
  - Remove HTML code
  - Handle emojis (convert to text or numerical values)
  - Use tools like demoji module in Python
- **Why quality matters:**
  - Newswires: formal, error-free
  - Social media: spelling errors, colloquialisms, abbreviations
  - Editing needed for most common issues

EXAM TRAP: Pre-processing converts raw text to analyzable format!

Q10: Document Similarity - Tutorial

QUESTION: What is a practical use of document similarity in finance?

OPTIONS:

1. To make documents longer
2. To assess whether a news article constitutes new information
3. To delete duplicate documents
4. To remove all documents

ANSWER: B

DETAILED TUTORIAL:

- **Document similarity applications:**
  1. **New information assessment:**
     - Determine if news article provides new information
     - Avoid redundant analysis
     - Focus on novel content
  2. **Disclosure tracking:**
     - Assess if tone of accounting disclosures has changed
     - Monitor consistency in communications
     - Detect significant changes
- **How it works:**
  - Compare document vectors
  - Compute similarity scores
  - Identify duplicates or near-duplicates
  - Track evolution of content

EXAM TRAP: Document similarity helps assess new information content!

Q11: Tokenization Definition - Tutorial

QUESTION: What is tokenization in NLP pre-processing?

OPTIONS:

1. Removing all punctuation from text
2. Separating a document into sentences and then into words
3. Converting all text to lowercase
4. Removing stop words

ANSWER: B

DETAILED TUTORIAL:

- **Tokenization process:**
  1. **Sentence tokenization:**
     - Split document at periods, question marks, exclamation marks
     - Creates sentence boundaries
  2. **Word tokenization:**
     - Split sentences into words
     - Remove punctuation, spacing, special symbols
     - Convert to lowercase
- **Importance:**
  - Creates individual text units for analysis
  - Foundation for all subsequent NLP steps
  - Enables counting and vectorization

EXAM TRAP: Tokenization = splitting into sentences and words!

Q12: Stop Words Removal - Tutorial

QUESTION: What are stop words and why are they removed?

OPTIONS:

1. Important words that convey meaning
2. Common words with no informational value (e.g., "the", "a", "also")
3. All nouns and verbs
4. All proper nouns

ANSWER: B

DETAILED TUTORIAL:

- **Stop words definition:**
  - Words that have little informational value
  - Included to make sentences flow
  - Examples: "has", "a", "the", "also", "and", "of"
  - Common in all documents
- **Why remove them:**
  - They add noise to analysis
  - They don't help distinguish documents
  - They increase vector size unnecessarily
  - Improve computational efficiency
- **Caveat:**
  - What constitutes a stop word varies by application
  - Words with no value in one context may be useful in another
  - Customization may be needed

EXAM TRAP: Stop words = common words with no informational value!

Q13: Stemming Definition - Tutorial

QUESTION: What is stemming in NLP pre-processing?

OPTIONS:

1. Removing all words from a document
2. Replacing words with their core/root forms by cutting prefixes and suffixes
3. Converting all text to uppercase
4. Removing all punctuation

ANSWER: B

DETAILED TUTORIAL:

- **Stemming definition:**
  - Reduces words to their root forms
  - Cuts off prefixes and suffixes
  - Crude, rule-based approach
- **Examples:**
  - "disappointing" $\rightarrow$ "disappoint"
  - "disappointed" $\rightarrow$ "disappoint"
  - "running" $\rightarrow$ "run"
  - "jumping" $\rightarrow$ "jump"
- **Purpose:**
  - Treat similar words equally in analysis
  - Reduce vocabulary size
  - Simplify analysis
- **Limitation:**
  - Can lose nuance and degrees of positivity/negativity
  - May create non-words
  - Less accurate than lemmatization

EXAM TRAP: Stemming = cutting prefixes/suffixes to root form!

Q14: Lemmatization Definition - Tutorial

QUESTION: What is lemmatization and how does it differ from stemming?

OPTIONS:

1. Lemmatization is the same as stemming
2. Lemmatization finds the dictionary form (lemma) considering word meaning and part of speech
3. Lemmatization removes stop words
4. Lemmatization converts text to lowercase

ANSWER: B

DETAILED TUTORIAL:

- **Lemmatization definition:**
  - Finds the dictionary form (lemma) of words
  - Considers word meaning and part of speech
  - More sophisticated than stemming
- **Comparison with stemming:**
  - **Stemming:**
    - Rule-based, cuts prefixes/suffixes
    - Fast, but can be inaccurate
    - May create non-words
  - **Lemmatization:**
    - Uses vocabulary and morphological analysis
    - Slower but more accurate
    - Always produces real words
- **Examples:**
  - "running" $\rightarrow$ "run" (lemma)
  - "better" $\rightarrow$ "good" (lemma)
  - "went" $\rightarrow$ "go" (lemma)

EXAM TRAP: Lemmatization = dictionary form considering meaning and POS!

Q15: POS Tagging Purpose - Tutorial

QUESTION: What is Part-of-Speech (POS) tagging and why is it useful?

OPTIONS:

1. Identifying all verbs in a document
2. Labelling each word by its class (noun, verb, adjective, proper noun, etc.)
3. Removing all adjectives from a document
4. Counting the number of sentences

ANSWER: B

DETAILED TUTORIAL:

- **POS tagging definition:**
  - Labels each word with its grammatical category
  - Common tags: noun, verb, adjective, proper noun, adverb
  - Provides syntactic information
- **Why it's useful:**
  - **Handles ambiguity:**
    - "Reading" (town) = proper noun
    - "reading" (activity) = verb
  - **Handles case sensitivity:**
    - "Polish" (language) = proper noun
    - "polish" (to shine) = verb
  - **Improves accuracy:**
    - Identifies proper nouns before lowercasing
    - Preserves important distinctions

EXAM TRAP: POS tagging labels each word by grammatical class!

Q16: Lowercasing Issues - Tutorial

QUESTION: What problem can lowercasing cause in NLP?

OPTIONS:

1. It makes text harder to read
2. It can cause inaccuracies by removing case-based distinctions (proper nouns, different meanings)
3. It doesn't affect NLP
4. It makes words longer

ANSWER: B

DETAILED TUTORIAL:

- **Lowercasing problems:**
  1. **Proper nouns:**
     - "Reading" (town) vs "reading" (activity)
     - "Polish" (language) vs "polish" (to shine)
     - Names of people, places, firms
  2. **Meaning loss:**
     - Important distinctions are lost
     - Can lead to misclassification
     - Reduces accuracy
- **Solution:**
  - Apply POS tagging before lowercasing
  - Preserve proper noun information
  - Maintain case-based distinctions

EXAM TRAP: Lowercasing can lose proper noun distinctions - use POS tagging first!

Q17: Punctuation Removal Effects - Tutorial

QUESTION: What is a limitation of removing punctuation?

OPTIONS:

1. It doesn't affect NLP
2. It can cause loss of valuable context (questions, exclamations, sarcasm)
3. It makes text shorter
4. It removes all verbs

ANSWER: B

DETAILED TUTORIAL:

- **Punctuation problems:**
  1. **Questions:**
     - "Is the CEO performing well?" $\rightarrow$ may seem positive
     - Without "?", classified as positive
     - Actually a question (little information content)
  2. **Exclamations:**
     - May denote humor or sarcasm
     - Different information value than period
     - Tone indication is lost
- **Solution:**
  - Consider punctuation separately
  - Identify question marks, exclamation points
  - Use as additional features
  - Context matters

EXAM TRAP: Removing punctuation loses context and tone information!

Q18: Stemming vs Lemmatization Usage - Tutorial

QUESTION: When might lemmatization be preferred over stemming?

OPTIONS:

1. When speed is the only concern
2. When accuracy and preserving meaning are important
3. When working with very short documents
4. When speed is critical

ANSWER: B

DETAILED TUTORIAL:

- **When to use lemmatization:**
  - **Accuracy is critical:**
    - Financial documents need precise analysis
    - Proper word forms matter
    - Avoid non-words
  - **Meaning preservation:**
    - Different forms of same word
    - Irregular verbs
    - Context-specific forms
- **When to use stemming:**
  - **Speed is critical:**
    - Very large documents
    - Real-time processing
    - Simpler analysis
- **Tradeoff:**
  - Stemming: faster, less accurate
  - Lemmatization: slower, more accurate

EXAM TRAP: Lemmatization preferred for accuracy, stemming for speed!

Q19: Vocabulary Building - Tutorial

QUESTION: How is the vocabulary for BoW determined?

OPTIONS:

1. Only using pre-existing dictionaries
2. Either imposed exogenously or created from every distinct word in the corpus
3. Only using the most common words
4. Only using the rarest words

ANSWER: B

DETAILED TUTORIAL:

- **Vocabulary creation methods:**
  1. **Exogenous creation:**
     - Created ahead of time
     - Applied to documents
     - Fixed and consistent
  2. **Corpus-based creation:**
     - Every distinct word in all documents
     - Comprehensive and complete
     - May be very large
- **Tradeoffs:**
  - **Exogenous:** Controlled but may miss domain-specific words
  - **Corpus-based:** Complete but may be too large
- **Processing:**
  - Remove very frequent words (top 5%)
  - Remove very rare words
  - Balance vocabulary size

EXAM TRAP: Vocabulary = pre-defined OR from corpus words!

Q20: Bag of Words Definition - Tutorial

QUESTION: What is the Bag of Words (BoW) approach?

OPTIONS:

1. Analyzing words in their original order
2. Treating each word in a document independently, ignoring order
3. Counting only the longest words
4. Removing all words from documents

ANSWER: B

DETAILED TUTORIAL:

- **BoW definition:**
  - Treats each word distinctly and equally
  - Position and order are ignored
  - Counts occurrences in each document
- **Key characteristics:**
  - **Independence:** Each word considered separately
  - **Equal treatment:** All words have same importance (in basic version)
  - **Simplification:** Removes complex language structure
- **Limitations:**
  - Loses context
  - Ignores negation words
  - Misses phrase meanings
  - n-grams can help address this

EXAM TRAP: BoW ignores word order and position!

Q21: Binary vs Count BoW - Tutorial

QUESTION: What is the difference between binary and count Bag of Words?

OPTIONS:

1. Binary uses 1/0, count uses frequency counts
2. Binary is more accurate than count
3. Count ignores word frequencies
4. There is no difference

ANSWER: A

DETAILED TUTORIAL:

- **Binary BoW:**
  - 1 if word appears in document
  - 0 if word does not appear
  - Presence/absence only
  - **Example:** [1, 0, 1, 0, ...]
- **Count BoW:**
  - Counts how many times each word appears
  - Non-negative integers
  - Frequency matters
  - **Example:** [2, 0, 1, 0, ...]
- **When to use each:**
  - **Binary:** When presence is enough
  - **Count:** When frequency matters
  - Usually count BoW gives more information

EXAM TRAP: Binary = presence/absence, Count = frequency counts!

Q22: Document Term Matrix - Tutorial

QUESTION: What is a Document Term Matrix (DTM)?

OPTIONS:

1. A matrix of document titles
2. A collection of document vectors with rows as documents and columns as terms
3. A matrix of document dates
4. A matrix of document authors

ANSWER: B

DETAILED TUTORIAL:

- **DTM definition:**
  - Rows = documents
  - Columns = terms/words
  - Entries = word counts or frequencies
  - Dimensions: $|W| \times D$
- **Characteristics:**
  - **Sparse:** Many zeros
  - **Large:** Many columns (vocabulary size)
  - **Quantitative:** Enables analysis
- **Example:**
  - 1,000 documents, 10,000 word vocabulary
  - Matrix: 1,000 $\times$ 10,000
  - Most entries are zero

EXAM TRAP: DTM = documents as rows, terms as columns!

Q23: Sparse Vectors Problem - Tutorial

QUESTION: Why are document word vectors described as sparse?

OPTIONS:

1. They contain many non-zero values
2. They contain many zeros
3. They are all the same length
4. They are very small

ANSWER: B

DETAILED TUTORIAL:

- **Sparsity definition:**
  - Most words don't appear in each document
  - Vector entries are mostly zero
  - Common in NLP due to large vocabularies
- **Example:**
  - Vocabulary: 10,000 words
  - Document: 100 unique words
  - 9,900 zeros in vector
  - 99% sparsity
- **Problems:**
  - **Computational inefficiency:**
    - Slow processing
    - Memory intensive
  - **Statistical issues:**
    - Curse of dimensionality
    - Hard to find patterns
- **Solutions:**
  - Dimensionality reduction
  - Feature selection
  - Sparse matrix representations

EXAM TRAP: Sparse = many zeros - makes modeling inefficient!

Q24: Vector Normalization - Tutorial

QUESTION: Why is vector normalization needed in NLP?

OPTIONS:

1. To make vectors longer
2. To prevent repeated words from skewing analysis
3. To remove all words
4. To make vectors shorter

ANSWER: B

DETAILED TUTORIAL:

- **Why normalize:**
  - Repeated words can skew analysis
  - Example: "Bad" repeated multiple times
  - Similar to outliers in econometrics
  - Need to equalize influence
- **L2 normalization:**
  - Divide each element by the square root of sum of squares
  - $||v||_2 = \sqrt{\sum x_i^2}$
  - Scales elements between 0 and 1
  - All documents become unit length
- **Example:**
  - Original: [2, 0, 1, 0, 1]
  - L2 norm: $2^2 + 0 + 1 + 0 + 1 = 6$
  - Normalized: $[2/\sqrt{6}, 0, 1/\sqrt{6}, 0, 1/\sqrt{6}]$
  - $\approx [0.82, 0, 0.41, 0, 0.41]$

EXAM TRAP: L2 normalization = scale to unit length!

Q25: L2 Norm Formula - Tutorial

QUESTION: What is the formula for the L2 norm of a vector?

OPTIONS:

1. $||X||_2 = \sum x_i$
2. $||X||_2 = \sqrt{\sum x_i^2}$
3. $||X||_2 = \sum x_i^2$
4. $||X||_2 = \max(x_i)$

ANSWER: B

DETAILED TUTORIAL:

- **L2 norm formula:**
  - $||X||_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$
  - Where x_i are elements of the vector
  - Also called Euclidean norm
- **Application in NLP:**
  - Normalize document vectors
  - Each vector becomes unit length
  - Comparable across documents
- **Example calculation:**
  - Vector: [2, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1]
  - Sum of squares $= 4 + 1 + 1 + 1 + 1 = 8$
  - L2 norm $= \sqrt{8} = 2.83$

EXAM TRAP: L2 norm = square root of sum of squared elements!

Q26: Dictionary Approach Definition - Tutorial

QUESTION: What is the dictionary approach in NLP?

OPTIONS:

1. A machine learning algorithm
2. A pre-defined list of words sharing a common characteristic used for text analysis
3. A method for removing stop words
4. A way to tokenize documents

ANSWER: B

DETAILED TUTORIAL:

- **Dictionary approach definition:**
  - Pre-defined list of words
  - Words share a common characteristic
  - Used to assess sentiment or topic
- **Common application:**
  - Sentiment analysis
  - Separate positive and negative words
  - Count occurrences in documents
- **Net sentiment score:**
  - Proportion positive - proportion negative
  - Divide each by total words
  - Neutral words can be ignored

EXAM TRAP: Dictionary = pre-defined word lists with common characteristics!

Q27: Dictionary Approach Example - Tutorial

QUESTION: In the newswire example, how is sentiment determined?

OPTIONS:

1. Count only positive words
2. Count only negative words
3. Count positive and negative words, subtract negatives from positives
4. Count all words equally

ANSWER: C

DETAILED TUTORIAL:

- **Sentiment calculation:**
  - Count positive words: rise, grow, resolve, relief (4 positives)
  - Count negative words: disappoint, worry, concern, decline (4 negatives)
  - Net sentiment = 4 - 4 = 0 (neutral)
- **Example text:**
  - "Firm XYZ reported rise in earnings, disappointing investors"
  - "Previous safety worries resolved, should underpin growth"
  - "Analyst expressed relief, concerns about decline"
- **Challenges:**
  - Counterfactual sentences ("would have led to decline")
  - Complex sentence structure
  - Negation words can reverse meaning

EXAM TRAP: Net sentiment = positive count - negative count!

Q28: Loughran-McDonald Dictionary - Tutorial

QUESTION: What is the Loughran-McDonald dictionary known for?

OPTIONS:

1. Being the largest dictionary
2. Being specifically designed for financial text analysis
3. Being the oldest dictionary
4. Only containing positive words

ANSWER: B

DETAILED TUTORIAL:

- **Loughran-McDonald dictionary:**
  - Developed for financial text analysis
  - Based on extensive examination of 10-K filings
  - Categories: positive, negative, uncertain, litigious, strong modal, weak modal
- **Accuracy:**
  - Around 75% accuracy on Thomson Reuters news articles
  - Widely used in finance research
- **Importance:**
  - General dictionaries misclassify financial text
  - Financial words have specific meanings
  - Example: "liability" is not negative in financial context

EXAM TRAP: Loughran-McDonald = financial text dictionary - 75% accuracy!

Q29: Dictionary Approach Advantages - Tutorial

QUESTION: What are the primary advantages of the dictionary approach?

OPTIONS:

1. It's always 100% accurate
2. Transparency, ease and speed of implementation
3. It works for all types of text
4. It requires no pre-processing

ANSWER: B

DETAILED TUTORIAL:

- **Key advantages:**
  1. **Transparency:**
     - Word lists are explicit and clear
     - Easy to understand what's being measured
     - Results are interpretable
  2. **Ease of implementation:**
     - Simple word counting
     - No training required
     - Fast to apply
  3. **Comparability:**
     - Same word lists across corpora
     - Consistent measurement
     - Enables comparison

EXAM TRAP: Advantages = transparency, ease, speed, comparability!

Q30: Dictionary Approach Limitations - Tutorial

QUESTION: What is a key limitation of the dictionary approach?

OPTIONS:

1. It's too fast
2. Dictionaries are only as good as the word lists they are built on
3. It always overestimates sentiment
4. It always underestimates sentiment

ANSWER: B

DETAILED TUTORIAL:

- **Key limitations:**
  1. **Quality dependence:**
     - Only as good as word lists
     - Incomplete lists $\rightarrow$ poor accuracy
     - Ambiguities $\rightarrow$ misclassification
     - Errors propagate
  2. **Context limitation:**
     - Designed for formal documents
     - Poor for social media
     - Colloquial language not captured
     - "Moon" not in financial dictionaries
  3. **Narrow scope:**
     - Limited subjects available
     - Time-consuming to create new dictionaries
     - Machine learning often preferred for new applications
  4. **Negation issues:**
     - "Not good" $\rightarrow$ should be negative
     - "Not bad" $\rightarrow$ should be positive
     - Negations away from objects are hard

EXAM TRAP: Dictionary = only as good as word lists - has many limitations!

Q31: n-gram Definition - Tutorial

QUESTION: What is an n-gram in NLP?

OPTIONS:

1. A single word
2. A sequence of n contiguous words treated as a single unit
3. A type of machine learning algorithm
4. A pre-processing technique

ANSWER: B

DETAILED TUTORIAL:

- **n-gram definition:**
  - n = 1: uni-gram (single word)
  - n = 2: bi-gram (two words)
  - n = 3: tri-gram (three words)
  - Treats contiguous words as single unit
- **Example:**
  - Document: "credit spreads narrowed today"
  - Uni-grams: ["credit", "spreads", "narrowed", "today"]
  - Bi-grams: ["credit spreads", "spreads narrowed", "narrowed today"]
- **Purpose:**
  - Capture phrase meanings
  - Handle negation words
  - Address context issues

EXAM TRAP: n-gram = n contiguous words as one unit!

Q32: Why Use n-grams? - Tutorial

QUESTION: Why are n-grams needed beyond basic BoW?

OPTIONS:

1. To make analysis faster
2. Because some words have specific meaning when placed together
3. Because basic BoW is too accurate
4. To count words more precisely

ANSWER: B

DETAILED TUTORIAL:

- **Limitations of basic BoW:**
  1. **Loses phrase meaning:**
     - "red herring" $\neq$ red + herring
     - "San Diego" $\neq$ San + Diego
     - "credit spread" $\neq$ credit + spread
  2. **Negation words:**
     - "not good" $\neq$ not + good
     - "nothing useful" $\neq$ nothing + useful
     - "neither up nor down" $\neq$ neither + up + nor + down
- **How n-grams help:**
  - Preserve phrase meaning
  - Handle negations correctly
  - Capture context
  - More accurate sentiment analysis

EXAM TRAP: n-grams capture phrase meaning and negation context!

Q33: Negation Handling with n-grams - Tutorial

QUESTION: How can n-grams help with negation words?

OPTIONS:

1. By removing all negation words
2. By treating "not" + following word as a single unit
3. By ignoring negations
4. By making negations positive

ANSWER: B

DETAILED TUTORIAL:

- **Negation problem:**
  - "not good" $\rightarrow$ should be negative sentiment
  - Basic BoW sees "not" and "good" separately
  - Misses the negation effect
- **n-gram solution:**
  - Bi-grams capture "not good" as one unit
  - "not sufficient" as one unit
  - "never great" as one unit
  - Preserves negation meaning
- **Example:**
  - "The service is not good"
  - Bi-grams: ["The service", "service is", "is not", "not good"]
  - "not good" detected as negative phrase

EXAM TRAP: n-grams pair negation + following word as one unit!

Q34: TF-IDF Definition - Tutorial

QUESTION: What does TF-IDF stand for and what does it measure?

OPTIONS:

1. Term Frequency - Inverse Document Frequency; weights words by importance
2. Total Frequency - Integrated Document Frequency; measures total occurrences
3. Text Frequency - Inverse Data Frequency; measures text length
4. Term Frequency - Inverse Data Frequency; measures document similarity

ANSWER: A

DETAILED TUTORIAL:

- **TF-IDF definition:**
  - **TF (Term Frequency):** How often word appears in document
  - **IDF (Inverse Document Frequency):** How rare the word is across documents
  - **TF-IDF:** TF $\times$ IDF = word importance score
- **Purpose:**
  - Assign higher weights to important words
  - Words frequent in document but rare in corpus = high score
  - Words frequent everywhere = low score
- **Intuition:**
  - Common words ("the", "this") get zero weight
  - Rare but relevant words get high weight
  - Better than simple word counts

EXAM TRAP: TF-IDF = Term Frequency $\times$ Inverse Document Frequency!

Q35: Term Frequency Formula - Tutorial

QUESTION: What is the formula for Term Frequency (TF)?

OPTIONS:

1. $TF_{i,j} = w_{i,j} \times |v_j|$
2. $TF_{i,j} = w_{i,j} / |v_j|$
3. $TF_{i,j} = |v_j| / w_{i,j}$
4. $TF_{i,j} = \log(w_{i,j})$

ANSWER: B

DETAILED TUTORIAL:

- **TF Formula:**
  - $TF_{i,j} = \frac{w_{i,j}}{|v_j|}$
  - $w_{i,j}$: number of times term i appears in document j
  - $|v_j|$: total number of terms in document j
  - Normalizes by document length
- **Why normalize:**
  - Longer documents have more words
  - Without normalization, longer documents dominate
  - Corrects for document length
- **Example:**
  - Word appears 3 times in 100-word document
  - TF = 3/100 = 0.03
  - Word appears 3 times in 50-word document
  - TF = 3/50 = 0.06 (higher importance)

EXAM TRAP: TF = word count / total words in document!

Q36: Inverse Document Frequency Formula - Tutorial

QUESTION: What is the formula for Inverse Document Frequency (IDF)?

OPTIONS:

1. $IDF_i = \ln(D / \sum d_{i,j})$
2. $IDF_i = D / \sum d_{i,j}$
3. $IDF_i = \sum d_{i,j} / D$
4. $IDF_i = \ln(\sum d_{i,j} / D)$

ANSWER: A

DETAILED TUTORIAL:

- **IDF Formula:**
  - $IDF_i = \ln(D / \sum_j d_{i,j})$
  - D = total number of documents
  - $\sum d_{i,j}$ = number of documents containing term i
  - Uses natural logarithm
- **Interpretation:**
  - Rare words: High IDF
  - Common words: Low IDF
  - Words in all documents: IDF = 0
- **Example:**
  - 100 documents, word appears in 10 documents
  - IDF = ln(100/10) = ln(10) = 2.30
  - Word appears in 90 documents
  - IDF = ln(100/90) = ln(1.11) = 0.10

EXAM TRAP: IDF = ln(total documents / documents containing word)!

Q37: Zero TF-IDF Value - Tutorial

QUESTION: Why do some words have zero TF-IDF values?

OPTIONS:

1. Because they are not in the vocabulary
2. Because they appear in every document (no discriminatory power)
3. Because they are too long
4. Because they are too short

ANSWER: B

DETAILED TUTORIAL:

- **Zero IDF explanation:**
  - IDF = ln(D / $\sum d_{i,j}$)
  - If word appears in all documents: $\sum d_{i,j} = D$
  - IDF = ln(D/D) = ln(1) = 0
  - TF-IDF = TF $\times$ 0 = 0
- **Why this matters:**
  - Word has no discriminatory power
  - Cannot distinguish between documents
  - No value for classification
- **Example:**
  - "this" appears in all 4 documents in Box 9.2
  - IDF = ln(4/4) = 0
  - TF-IDF = 0

EXAM TRAP: Zero TF-IDF = word appears in all documents - no discriminatory power!

Q38: TF-IDF Variation by Document - Tutorial

QUESTION: Why does the same word have different TF-IDF values across documents?

OPTIONS:

1. IDF varies by document
2. TF varies by document (due to different lengths and frequencies)
3. TF-IDF is always constant
4. Vocabulary changes

ANSWER: B

DETAILED TUTORIAL:

- **Why TF varies:**
  1. **Document length:**
     - TF depends on document length
     - Longer documents have smaller TFs
     - Same word count, different TF
  2. **Frequency:**
     - Word may appear multiple times in one document
     - Once in another document
     - TF changes accordingly
- **IDF is constant:**
  - IDF depends only on corpus
  - Same for all documents
  - TF varies by document
  - TF-IDF = TF $\times$ IDF $\Rightarrow$ varies by document

EXAM TRAP: IDF is constant across documents; TF varies $\Rightarrow$ TF-IDF varies!

Q39: TF-IDF Variations - Tutorial

QUESTION: What are common variations in TF-IDF implementation?

OPTIONS:

1. Only one formula exists
2. Variations include binary TF, log normalization, and different IDF formulas
3. TF-IDF is always the same
4. TF-IDF cannot be modified

ANSWER: B

DETAILED TUTORIAL:

- **Common variations:**
  1. **TF variations:**
     - Binary TF: 1 if word appears, 0 otherwise
     - Raw count: $w_{i,j}$ (no normalization)
     - Log normalization: $\ln(1 + w_{i,j})$
     - Normalized: $w_{i,j}/|v_j|$ (standard)
  2. **IDF variations:**
     - Smooth IDF: $\ln((D+1)/(\sum d_{i,j}+1)) + 1$
     - Standard IDF: $\ln(D/\sum d_{i,j})$
     - Probabilistic IDF: $\log((D - \sum d_{i,j})/\sum d_{i,j})$
- **Why variations exist:**
  - Different applications need different emphasis
  - Some handle document length better
  - Some are more robust
  - Different software implementations

EXAM TRAP: TF-IDF has many implementation variations - be aware!

Q40: TF-IDF Example Interpretation - Tutorial

QUESTION: In the TF-IDF example, which word has the highest score and why?

OPTIONS:

1. "this" - because it appears in all documents
2. "fed" - because it appears in only one short document
3. "down" - because it appears frequently
4. "stock" - because it appears in two documents

ANSWER: B

DETAILED TUTORIAL:

- **Highest scores:**
  - "fed" = 0.28 (TF-IDF)
  - "hikes" = 0.28 (TF-IDF)
  - "rates" = 0.28 (TF-IDF)
- **Why they're highest:**
  - Appear in only 1 out of 4 documents
  - IDF = $\ln(4/1) = 1.39$ (high)
  - Document length = 5 (short)
  - TF = $1/5 = 0.20$
  - TF-IDF = $0.20 \times 1.39 = 0.28$
- **Lowest scores:**
  - "this" = 0 (appears in all documents)
  - "down" = 0.05 (appears in 3 of 4 documents)

EXAM TRAP: Rare words in short documents = highest TF-IDF!

Q41: Naïve Bayes Definition - Tutorial

QUESTION: What is the Naïve Bayes classifier?

OPTIONS:

1. A regression algorithm
2. A classification algorithm based on Bayes' theorem with independence assumption
3. A clustering algorithm
4. A dimensionality reduction technique

ANSWER: B

DETAILED TUTORIAL:

- **Naïve Bayes definition:**
  - Based on Bayes' theorem
  - Assumes features are independent (naïve assumption)
  - Used for classification tasks
  - Popular for text classification
- **Why "naïve":**
  - Assumes word independence
  - In reality, words are not independent
  - Still works well in practice
- **Advantages:**
  - Simple and fast
  - Solid classification accuracy
  - Works well with high-dimensional data
  - Easy to implement

EXAM TRAP: Naïve = independence assumption - words treated independently!

Q42: Bayes' Theorem - Tutorial

QUESTION: What is Bayes' theorem in the context of classification?

OPTIONS:

1. $P(c|d) = P(d|c)P(c)/P(d)$
2. $P(c|d) = P(d|c) + P(c)$
3. $P(c|d) = P(c)/P(d)$
4. $P(c|d) = P(d)/P(c)$

ANSWER: A

DETAILED TUTORIAL:

- **Bayes' Theorem:**
  - $P(c|d) = \frac{P(d|c) \times P(c)}{P(d)}$
  - $P(c|d)$: Posterior probability (probability of class given document)
  - $P(d|c)$: Likelihood (probability of document given class)
  - $P(c)$: Prior probability (unconditional probability of class)
  - $P(d)$: Predictor prior probability (normalizing constant)
- **Application:**
  - Calculate probability for each class
  - Assign document to class with highest probability
  - $P(d)$ cancels across classes (can be ignored)

EXAM TRAP: Bayes = posterior = likelihood $\times$ prior / evidence!

Q43: Naïve Independence Assumption - Tutorial

QUESTION: What is the independence assumption in Naïve Bayes?

OPTIONS:

1. Documents are independent of each other
2. Each word in a document is independent of all other words
3. Classes are independent of documents
4. All documents have the same length

ANSWER: B

DETAILED TUTORIAL:

- **Independence assumption:**
  - $P(w_1, w_2, ..., w_n|c) = P(w_1|c) \times P(w_2|c) \times ... \times P(w_n|c)$
  - Words are assumed conditionally independent
  - Given the class, word probabilities multiply
- **Why it's naïve:**
  - In reality, words are not independent
  - "good" and "great" often appear together
  - "not" and "good" are related
- **Why it works:**
  - Despite unrealistic assumption
  - Works well in practice
  - Simple and effective

EXAM TRAP: Independence = words treated as independent - unrealistic but works!

Q44: Prior Probability - Tutorial

QUESTION: How is prior probability $P(c)$ calculated in Naïve Bayes?

OPTIONS:

1. $P(c) = n(c)/N$ (documents in class / total documents)
2. $P(c) = N/n(c)$
3. $P(c) = n(c) \times N$
4. $P(c) = 1/N$

ANSWER: A

DETAILED TUTORIAL:

- **Prior probability formula:**
  - $P(c_i) = \frac{n(c_i)}{N}$
  - $n(c_i)$: number of documents labeled as class i
  - N: total number of documents
  - Unconditional probability of each class
- **Example:**
  - 8 total documents
  - 4 good, 2 bad, 2 indifferent
  - $P(good) = 4/8 = 0.5$
  - $P(bad) = 2/8 = 0.25$
  - $P(indifferent) = 2/8 = 0.25$
- **Importance:**
  - Prior knowledge about class frequencies
  - Updates to posterior probability
  - Helps when likelihoods are similar

EXAM TRAP: Prior = class count / total documents!

Q45: Likelihood Calculation - Tutorial

QUESTION: How is the likelihood $P(w|c)$ calculated in Naïve Bayes?

OPTIONS:

1. Count of word occurrences in class / total words in class
2. Count of documents in class / total documents
3. Word count / total words in corpus
4. Always 0.5

ANSWER: A

DETAILED TUTORIAL:

- **Likelihood calculation:**
  - $P(w_i|c_j) = \frac{\text{count of } w_i \text{ in class } c_j}{\text{total words in class } c_j}$
  - Conditional probability of word given class
  - Estimated from labeled training data
- **Example from customer service:**
  - Class 1 (good): 4 documents
  - Word 2 appears in 1 of 4 good documents
  - $P(w_2|good) = 1/4 = 0.25$
- **Important:**
  - Zero probabilities are problematic
  - Need smoothing to avoid zeros

EXAM TRAP: Likelihood = word count in class / total words in class!

Q46: Posterior Probability - Tutorial

QUESTION: What is the posterior probability in Naïve Bayes?

OPTIONS:

1. The probability of a document given its class
2. The probability of a class given the document
3. The probability of a word given its class
4. The unconditional class probability

ANSWER: B

DETAILED TUTORIAL:

- **Posterior definition:**
  - $P(c|d)$
  - Probability that document d belongs to class c
  - Updated after seeing the document
  - Combines prior and likelihood
- **Calculation:**
  - $P(c|d) \propto P(d|c) \times P(c)$
  - Compute for each class
  - Choose class with highest posterior
- **Example:**
  - After seeing words 1 and 2
  - $P(bad|words) = 0.67$
  - $P(indifferent|words) = 0.33$
  - $P(good|words) = 0$
  - Classify as "bad"

EXAM TRAP: Posterior = probability of class given document!

Q47: Maximum A Posteriori - Tutorial

QUESTION: What is Maximum A Posteriori (MAP) classification?

OPTIONS:

1. Assign document to the class with the highest posterior probability
2. Assign document to the class with the highest prior probability
3. Assign document to the class with the highest likelihood
4. Random classification

ANSWER: A

DETAILED TUTORIAL:

- **MAP definition:**
  - $\hat{c} = \arg\max_{c \in C} P(c|d)$
  - Choose class that maximizes posterior
  - "Maximum A Posteriori"
- **Simplified form:**
  - $\hat{c} = \arg\max_{c \in C} P(d|c) \times P(c)$
  - $P(d)$ cancels out
  - Only need likelihood and prior
- **Example:**
  - $P(bad|words) = 0.67$
  - $P(indifferent|words) = 0.33$
  - $P(good|words) = 0$
  - MAP class = "bad"

EXAM TRAP: MAP = class with highest posterior probability!

Q48: Zero Probability Problem - Tutorial

QUESTION: What is the zero probability problem in Naïve Bayes?

OPTIONS:

1. When a word appears in all documents
2. When a word does not appear in a class in the training data, making the class probability zero
3. When all words have zero frequency
4. When no documents exist

ANSWER: B

DETAILED TUTORIAL:

- **Zero probability problem:**
  - Word doesn't appear in training data for a class
  - $P(w|c) = 0$
  - Product becomes zero: $\prod P(w_i|c) = 0$
  - Class probability becomes zero
- **Example:**
  - No "good" reviews contain word "slow"
  - New document has "slow" and otherwise positive words
  - Probability of "good" $= 0$
  - But review could still be good!
- **Consequences:**
  - Overly draconian
  - Unrealistic zero probabilities
  - Can misclassify documents

EXAM TRAP: Zero probability = word missing in training class - need smoothing!

Q49: Laplace Smoothing - Tutorial

QUESTION: What is Laplace smoothing in Naïve Bayes?

OPTIONS:

1. A method to remove stop words
2. Adding 1 to all counts to avoid zero probabilities
3. A technique to stem words
4. A method to normalize vectors

ANSWER: B

DETAILED TUTORIAL:

- **Laplace smoothing:**
  - Adds 1 (or λ) to all counts
  - Ensures all probabilities are non-zero
  - $P(w|c) = \frac{\text{count}(w,c)+1}{\text{total words in c}+|V|}$
- **Effect:**
  - Zero probabilities become positive
  - All probabilities pushed toward uniform
  - Larger λ = stronger smoothing
  - Impact decreases with sample size
- **Example:**
  - Before: $P(w|c) = 0/4 = 0$
  - After: $P(w|c) = 1/6 = 0.167$
  - Probability increases from 0 to small positive value

EXAM TRAP: Laplace smoothing = add 1 to all counts to avoid zeros!

Q50: Smoothing Impact on Probabilities - Tutorial

QUESTION: How does smoothing affect the estimated probabilities?

OPTIONS:

1. It makes probabilities more extreme
2. It pushes probabilities toward a uniform distribution
3. It has no effect on probabilities
4. It makes probabilities negative

ANSWER: B

DETAILED TUTORIAL:

- **Effect of smoothing:**
  - Prevents zero probabilities
  - Shrinks probabilities toward uniform
  - Reduces extreme values
  - Similar to regularization (Lasso/Ridge)
- **Impact of λ:**
  - $\lambda = 0$: No smoothing (original)
  - $\lambda = 1$: Laplace smoothing (common)
  - $\lambda > 1$: Stronger smoothing
  - Larger λ = more uniform probabilities
- **Sample size effect:**
  - Small sample: Smoothing has large impact
  - Large sample: Smoothing has small impact
  - Effect diminishes with more data

EXAM TRAP: Smoothing = shrinks toward uniform distribution - similar to regularization!

Q51: Underflow Problem - Tutorial

QUESTION: What is the underflow problem in Naïve Bayes?

OPTIONS:

1. When probabilities become too large
2. When multiplying many small probabilities results in numbers too small for computers to handle
3. When documents are too long
4. When vocabulary is too small

ANSWER: B

DETAILED TUTORIAL:

- **Underflow definition:**
  - Multiplying many probabilities < 1
  - Product becomes extremely small
  - Below computer precision
  - Rounds to zero
- **Cause:**
  - $P(w_1|c) \times P(w_2|c) \times \ldots \times P(w_n|c)$
  - Each term < 1
  - For long documents, product $\to 0$
  - Example: $0.5^{100} \approx 7.9 \times 10^{-31}$
- **Solution:**
  - Take logarithms
  - $\log(P(w_1|c) \times \ldots) = \sum \log(P(w_i|c))$
  - Sums instead of products
  - Computationally stable

EXAM TRAP: Underflow = multiplying small probabilities - use logarithms!

Q52: Log Transformation - Tutorial

QUESTION: Why is log transformation used in Naïve Bayes?

OPTIONS:

1. To make probabilities larger
2. To convert products to sums and avoid underflow
3. To remove stop words
4. To normalize the data

ANSWER: B

DETAILED TUTORIAL:

- **Log transformation benefits:**
  1. **Products to sums:**
     - $\log(P_1 \times P_2 \times \ldots) = \log(P_1) + \log(P_2) + \ldots$
     - Avoids multiplication of small numbers
     - Computationally stable
  2. **Preserves ordering:**
     - Log is monotonic
     - Class with highest probability = class with highest log probability
     - Results unchanged
  3. **Simplifies calculations:**
     - Sums are easier to compute
     - Less risk of numerical issues
     - More efficient
- **Example:**
  - Instead of: $0.5 \times 0.25 \times 0.33$
  - Use: $\log(0.5) + \log(0.25) + \log(0.33)$

EXAM TRAP: Log = products to sums - avoids underflow!

Q53: Unknown Words Handling - Tutorial

QUESTION: How are words in test data that were not in training data handled?

OPTIONS:

1. They are added to the vocabulary
2. They are ignored because they cannot help classify the document
3. They cause the classification to fail
4. They are treated as stop words

ANSWER: B

DETAILED TUTORIAL:

- **Unknown word handling:**
  - Words not in training vocabulary
  - No information about their distribution
  - Cannot help with classification
  - Must be removed from test document
- **Why ignore them:**
  - $P(w|c)$ unknown
  - Cannot calculate likelihood
  - Would cause problems
  - Better to exclude them
- **Solution:**
  - Build vocabulary from training data
  - Only use known words for classification
  - Unknown words provide no discriminatory power

EXAM TRAP: Unknown words = ignored - they provide no classification information!

Q54: Naïve Bayes Assumptions - Tutorial

QUESTION: What are the two key assumptions of Naïve Bayes for document classification?

OPTIONS:

1. Words are dependent and have different weights
2. Words are independent and have equal weighting
3. Words are ordered and have different weights
4. Words are random and have no weights

ANSWER: B

DETAILED TUTORIAL:

- **Assumption 1: Independence**
  - Each word is independent of others
  - $P(w_1, w_2, \ldots, w_n|c) = \prod P(w_i|c)$
  - Simplifies calculations
- **Assumption 2: Equal weighting**
  - Each word has equal information value
  - No word is more important than others
  - After stop word removal
- **Why these assumptions:**
  - Make problem tractable
  - Work surprisingly well
  - Simple and fast

EXAM TRAP: Naïve Bayes = independence + equal weighting!

Q55: Generative vs Discriminative - Tutorial

QUESTION: What makes Naïve Bayes a generative classifier?

OPTIONS:

1. It can generate new documents
2. It learns the joint distribution $P(d, c)$ and can create new classifications
3. It only learns decision boundaries
4. It doesn't use probabilities

ANSWER: B

DETAILED TUTORIAL:

- **Generative classifiers:**
  - Model the joint distribution $P(d, c)$
  - Learn how data is generated
  - Can generate new examples
  - Examples: Naïve Bayes, Hidden Markov Models
- **Discriminative classifiers:**
  - Model the decision boundary $P(c|d)$ directly
  - Only care about classification
  - Examples: Logistic regression, SVM
- **Difference:**
  - Generative: Complete model of data
  - Discriminative: Only classification boundary
  - Generative can create new classifications

EXAM TRAP: Naïve Bayes = generative - learns joint distribution!

Q56: Customer Service Example - Tutorial

QUESTION: In the customer service example, what classification is made and why?

OPTIONS:

1. "Good" because it has the highest prior
2. "Bad" because it has the highest posterior probability
3. "Indifferent" because it's the middle class
4. Cannot be classified

ANSWER: B

DETAILED TUTORIAL:

- **Data:**
  - 4 good, 2 bad, 2 indifferent (8 total)
  - New document: words 1 and 2 present, words 3 and 4 absent
- **Calculated probabilities:**
  - $P(good|words) = 0$ (without smoothing)
  - $P(bad|words) = 0.67$ (highest)
  - $P(indifferent|words) = 0.33$
- **Classification:**
  - "Bad" because 0.67 ¿ 0.33 ¿ 0
  - Even though prior for "good" was highest (0.5)
  - Likelihood updates prior to posterior

EXAM TRAP: Classification = highest posterior probability - not just prior!

Q57: Smoothing Effect Example - Tutorial

QUESTION: How did Laplace smoothing change the classification in the customer service example?

OPTIONS:

1. It didn't change the classification
2. It changed "Good" from 0 to 2.9% probability
3. It made "Bad" the lowest probability
4. It made classification impossible

ANSWER: B

DETAILED TUTORIAL:

- **Before smoothing:**
  - $P(good|words) = 0$
  - $P(bad|words) = 0.67$
  - $P(indifferent|words) = 0.33$
- **After smoothing ($\lambda = 1$):**
  - $P(good|words) = 0.029$ (increased from 0)
  - $P(bad|words) = 0.583$ (decreased)
  - $P(indifferent|words) = 0.388$ (increased slightly)
- **Classification unchanged:**
  - "Bad" still highest (0.583 ¿ 0.388 ¿ 0.029)
  - But probability of "Good" no longer zero
  - More realistic estimates

EXAM TRAP: Smoothing eliminates zero probabilities but may not change classification!

Q58: Multi-class Classification - Tutorial

QUESTION: How does Naïve Bayes handle more than two classes?

OPTIONS:

1. It cannot handle more than two classes
2. It computes posterior probability for each class and selects the highest
3. It only uses binary classification
4. It combines classes

ANSWER: B

DETAILED TUTORIAL:

- **Multi-class extension:**
  - Natural extension of binary case
  - Compute $P(c|d)$ for each class
  - Select class with highest probability
  - No limitation on number of classes
- **Example (3 classes):**
  - $P(good|d) = 0.029$
  - $P(bad|d) = 0.583$
  - $P(indifferent|d) = 0.388$
  - Select "bad" (highest probability)
- **Applications:**
  - Sentiment: positive, negative, neutral
  - Topic classification: finance, politics, sports, etc.
  - Document categorization

EXAM TRAP: Multi-class = choose class with highest posterior!

Q59: Naïve Bayes Advantages - Tutorial

QUESTION: What are the main advantages of Naïve Bayes?

OPTIONS:

1. It always has 100% accuracy
2. Simplicity, speed, and solid classification accuracy
3. It works without any assumptions
4. It requires no training data

ANSWER: B

DETAILED TUTORIAL:

- **Key advantages:**
  1. **Simplicity:**
     - Easy to understand
     - Easy to implement
     - Few parameters to tune
  2. **Speed:**
     - Fast training
     - Fast prediction
     - Scales to large datasets
  3. **Accuracy:**
     - Solid classification accuracy
     - Works well despite assumptions
     - Competitive with more complex models
- **When to use:**
  - High-dimensional data
  - Text classification
  - Quick baseline models

EXAM TRAP: Naïve Bayes advantages = simple, fast, and accurate!

Q60: Appendix Problem - Tutorial

QUESTION: In the appendix problem, what classification is made for the new instance?

OPTIONS:

1. Buy (probability = 0.98)
2. Sell (probability = 0.02)
3. Cannot be classified
4. Both equally likely

ANSWER: A

DETAILED TUTORIAL:

- **Data:**
  - 5 buys, 5 sells
  - Prior: $P(buy) = 0.5$, $P(sell) = 0.5$
  - New document: words 1, 2, 3 present; words 4, 5 absent
- **Conditional probabilities:**
  - $P(buy) \times \prod P(w_i|buy) = 0.0768$
  - $P(sell) \times \prod P(w_i|sell) = 0.00192$
- **Posterior probabilities:**
  - $P(buy|words) = 0.0768/(0.0768 + 0.00192) = 0.98$
  - $P(sell|words) = 0.00192/(0.0768 + 0.00192) = 0.02$
- **Classification:** Buy

EXAM TRAP: Posterior normalization = divide by sum of all posteriors!

Q61: Word Embedding Definition - Tutorial

QUESTION: What is word embedding in NLP?

OPTIONS:

1. A method to remove words
2. A technique to capture word meaning by representing words in a continuous vector space
3. A method to count word frequencies
4. A technique to stem words

ANSWER: B

DETAILED TUTORIAL:

- **Word embedding definition:**
  - Represents words as dense vectors
  - Captures semantic meaning
  - Similar words have similar vectors
  - Based on distributional hypothesis
- **Distributional hypothesis:**
  - "You shall know a word by the company it keeps"
  - Similar meanings appear in similar contexts
  - Words appearing together are related
- **Advantages over BoW:**
  - Captures meaning
  - Dense vectors (not sparse)
  - Similar words close in space
  - Better for downstream tasks

EXAM TRAP: Word embeddings = capture meaning in dense vectors!

Q62: Word2Vec Definition - Tutorial

QUESTION: What is Word2Vec?

OPTIONS:

1. A word stemming algorithm
2. A Google-developed algorithm for generating word embeddings using neural networks
3. A dictionary approach to NLP
4. A pre-processing technique

ANSWER: B

DETAILED TUTORIAL:

- **Word2Vec definition:**
  - Developed by Google (Mikolov et al., 2013)
  - Neural network-based algorithm
  - Generates word embeddings
  - Reduces dimensionality
- **Two architectures:**
  1. **CBOW (Continuous Bag of Words):**
     - Predict target word from context
     - Faster to train
  2. **Skip-gram:**
     - Predict context words from target word
     - Better for rare words
     - More computationally expensive
- **Capabilities:**
  - Captures semantic relationships
  - "king - man + woman equals queen"
  - Semantic analogies

EXAM TRAP: Word2Vec equal Google algorithm for word embeddings!

Q63: Context in Word Embeddings - Tutorial

QUESTION: How do word embeddings capture context?

OPTIONS:

1. By ignoring context
2. By paying attention to positions of words and their surrounding words
3. By counting all words equally
4. By removing all punctuation

ANSWER: B

DETAILED TUTORIAL:

- **Context capture methods:**
  1. **Position information:**
     - Track which words appear together
     - Window-based co-occurrence
     - Similar words appear in similar positions
  2. **Surrounding words:**
     - Consider words before and after
     - Skip-gram uses surrounding context
     - CBOW uses context to predict target
- **One-hot encoding limitation:**
  - Large and sparse
  - No notion of similarity
  - Each word independent
- **Embedding solution:**
  - Dense vectors
  - Similar contexts = similar vectors
  - Captures semantic meaning

EXAM TRAP: Word embeddings capture context by analyzing surrounding words!

Q64: Skip-gram vs CBOW - Tutorial

QUESTION: What is the difference between Skip-gram and CBOW?

OPTIONS:

1. They are the same algorithm
2. Skip-gram predicts context from target; CBOW predicts target from context
3. Skip-gram is for English only
4. CBOW is for English only

ANSWER: B

DETAILED TUTORIAL:

- **CBOW (Continuous Bag of Words):**
  - Input: context words
  - Output: target word
  - Faster to train
  - Better for frequent words
  - "Given surrounding words, what's the missing word?"
- **Skip-gram:**
  - Input: target word
  - Output: context words
  - Better for rare words
  - More computationally expensive
  - "Given a word, what words surround it?"
- **Choice depends on:**
  - Data size
  - Word frequency distribution
  - Computational resources

EXAM TRAP: Skip-gram = target $\rightarrow$ context; CBOW = context $\rightarrow$ target!

Q65: One-Hot Encoding - Tutorial

QUESTION: What is one-hot encoding in NLP?

OPTIONS:

1. A technique to encode temperature
2. Representing words as binary vectors with a 1 in the position of that word and 0 elsewhere
3. A method to count word frequencies
4. A technique to remove stop words

ANSWER: B

DETAILED TUTORIAL:

- **One-hot encoding:**
  - Vector length = vocabulary size
  - One position = 1 (the word's position)
  - All other positions = 0
  - Example: [0, 0, 1, 0, 0, ...]
- **Limitations:**
  - Very sparse
  - No semantic meaning
  - All words equidistant
  - Cannot capture relationships
- **Comparison with embeddings:**
  - One-hot: Sparse, no meaning
  - Embeddings: Dense, captures meaning

EXAM TRAP: One-hot = binary vector with one 1 - no semantic meaning!

Q66: Dimensionality Reduction in NLP - Tutorial

QUESTION: Why is dimensionality reduction important in NLP?

OPTIONS:

1. To make documents longer
2. To reduce sparse, high-dimensional vectors to dense, lower-dimensional representations
3. To add more features
4. To increase computation time

ANSWER: B

DETAILED TUTORIAL:

- **The problem:**
  - Vocabulary can have 10,000+ words
  - Vectors are very long
  - Very sparse (many zeros)
  - Slow and inefficient
- **Reduction methods:**
  1. **PCA:** Principal Component Analysis
  2. **SVD:** Singular Value Decomposition
  3. **Word2Vec:** Neural network-based
  4. **t-SNE:** For visualization
- **Benefits:**
  - Dense representations
  - Faster computation
  - Better performance
  - Captures relationships

EXAM TRAP: Dimensionality reduction = sparse $\rightarrow$ dense representations!

Q67: BoW vs Word Embeddings - Tutorial

QUESTION: What is a key advantage of word embeddings over BoW?

OPTIONS:

1. Embeddings are sparser
2. Embeddings capture semantic meaning and relationships between words
3. Embeddings are faster to compute
4. Embeddings don't require pre-processing

ANSWER: B

DETAILED TUTORIAL:

- **BoW limitations:**
  - Loses context
  - No semantic meaning
  - Sparse vectors
  - Ignores word relationships
- **Embedding advantages:**
  1. **Semantic meaning:**
     - Similar words close in space
     - Captures relationships
     - Analogies possible
  2. **Dense vectors:**
     - Lower dimensional
     - More efficient
     - Better for downstream tasks
  3. **Context awareness:**
     - Considers surrounding words
     - Better representation
     - More accurate

EXAM TRAP: Embeddings = capture semantic meaning - BoW doesn't!

Q68: Skip-gram Details - Tutorial

QUESTION: What is Skip-gram used for in NLP?

OPTIONS:

1. To remove stop words
2. To predict surrounding words given a target word
3. To count word frequencies
4. To stem words

ANSWER: B

DETAILED TUTORIAL:

- **Skip-gram architecture:**
  - Input: target word
  - Output: context words (surrounding words)
  - Uses neural network
  - Predicts probability of context words
- **Training objective:**
  - Maximize probability of context words
  - Given target word
  - Slide window over text
  - Learn embeddings
- **Window size:**
  - Typically 2-5 words each side
  - Larger window = more global context
  - Smaller window = more syntactic

EXAM TRAP: Skip-gram = target word $\rightarrow$ predict context words!

Q69: Word2Vec Application - Tutorial

QUESTION: What is a famous semantic relationship captured by Word2Vec?

OPTIONS:

1. "king - man + woman equals queen"
2. "king + man + woman equals queen"
3. "king - queen equals man - woman"
4. All words are equally related

ANSWER: A

DETAILED TUTORIAL:

- **Famous analogy:**
  - $v_{king} - v_{man} + v_{woman} \approx v_{queen}$
  - Captures gender relationship
  - Preserves arithmetic operations
- **Other examples:**
  - "Paris - France + Italy equals Rome"
  - "walking - walk + run equals running"
  - Captures multiple relationships
- **Why it works:**
  - Vector space captures semantic relationships
  - Similar relationships have similar vector offsets
  - Enables analogical reasoning

EXAM TRAP: Word2Vec captures semantic relationships via vector arithmetic!

Q70: NLP Evaluation Overview - Tutorial

QUESTION: How are NLP models evaluated?

OPTIONS:

1. Only by accuracy
2. Using confusion matrices, accuracy, precision, recall, F1 when labels exist
3. Only by speed
4. Only by data requirements

ANSWER: B

DETAILED TUTORIAL:

- **Labeled data evaluation:**
  1. **Confusion Matrix:** TP, TN, FP, FN
  2. **Accuracy:** Overall correctness
  3. **Precision:** Of predicted positives, how many were right?
  4. **Recall:** Of actual positives, how many did we catch?
  5. **F1:** Harmonic mean of precision and recall
- **Unlabeled data evaluation:**
  - Use unsupervised evaluation methods
  - Qualitative assessment
  - Subject matter expert review
- **Other considerations:**
  - Speed of algorithm
  - Data requirements
  - Computational resources

EXAM TRAP: Evaluation = confusion matrix metrics when labels exist!

Q71: NLP Model Evaluation Metrics - Tutorial

QUESTION: What metrics are used to evaluate NLP classification models?

OPTIONS:

1. Only accuracy
2. Accuracy, precision, recall, and F1 from confusion matrix
3. Only precision
4. Only recall

ANSWER: B

DETAILED TUTORIAL:

- **Standard evaluation metrics:**
  1. **Accuracy:**
     - $(TP + TN)/(TP + TN + FP + FN)$
     - Overall correctness
  2. **Precision:**
     - $TP/(TP + FP)$
     - "Of documents predicted as class X, how many were correct?"
  3. **Recall:**
     - $TP/(TP + FN)$
     - "Of documents actually in class X, how many did we catch?"
  4. **F1:**
     - Harmonic mean of precision and recall
     - Balances both metrics

EXAM TRAP: Use multiple metrics - not just accuracy!

Q72: Labeled vs Unlabeled Data - Tutorial

QUESTION: How is NLP evaluation different for unlabeled data?

OPTIONS:

1. No evaluation is possible
2. Use unsupervised evaluation methods or qualitative assessment
3. Use the same metrics as labeled data
4. Evaluation is not needed

ANSWER: B

DETAILED TUTORIAL:

- **Unlabeled data challenges:**
  - No "right answer" to compare against
  - Cannot compute confusion matrix
  - Need different evaluation approaches
- **Evaluation methods:**
  1. **Subject matter experts:**
     - Manually review sample of classifications
     - Assess accuracy qualitatively
     - Expensive but reliable
  2. **Unsupervised metrics:**
     - Cohesion and separation
     - Silhouette score
     - Domain-specific validation
  3. **Indirect validation:**
     - Check if results make sense
     - Correlation with known variables
     - Predictive power for other tasks

EXAM TRAP: Unlabeled data requires alternative evaluation methods!

Q73: Speed and Data Requirements - Tutorial

QUESTION: Why should speed and data requirements be considered in NLP evaluation?

OPTIONS:

1. They are not important
2. Textual databases can be vast and sparse vectors are hard to handle efficiently
3. Speed is never a concern
4. All NLP models are equally fast

ANSWER: B

DETAILED TUTORIAL:

- **Practical considerations:**
  1. **Data volume:**
     - NLP datasets can be enormous
     - Millions of documents
     - Billions of words
  2. **Sparsity:**
     - Sparse vectors are inefficient
     - Slow to process
     - Memory intensive
  3. **Model complexity:**
     - Neural networks are slow to train
     - Require large amounts of data
     - May be impractical for some applications
- **Tradeoffs:**
  - Accuracy vs speed
  - Complexity vs feasibility
  - Resource constraints matter

EXAM TRAP: NLP can be computationally expensive - speed and data matter!

Q74: BoW Limitations - Tutorial

QUESTION: What is a major limitation of simple BoW methods?

OPTIONS:

1. They are too fast
2. They are not useful for determining context or intended meaning
3. They are too accurate
4. They require too much data

ANSWER: B

DETAILED TUTORIAL:

- **BoW limitations:**
  1. **Loss of context:**
     - Word order ignored
     - Phrase meaning lost
     - No understanding of relationships
  2. **No semantic meaning:**
     - Words independent
     - No concept of similarity
     - "good" and "great" treated equally
  3. **Negation issues:**
     - "not good" vs "good" both contain "good"
     - Sentiment is reversed but BoW doesn't know
- **Advanced solutions:**
  - n-grams
  - Skip-grams
  - Word embeddings
  - Contextual embeddings (BERT)

EXAM TRAP: BoW fails to capture context and meaning!

Q75: Advanced NLP Techniques - Tutorial

QUESTION: What advanced techniques help determine word context?

OPTIONS:

1. Only BoW
2. n-grams, Skip-grams, and word embeddings
3. Only stop word removal
4. Only stemming

ANSWER: B

DETAILED TUTORIAL:

- **Context-aware techniques:**
  1. **n-grams:**
     - n contiguous words as one unit
     - Captures phrase meaning
     - Handles negations
  2. **Skip-grams:**
     - Predict context from target
     - Captures semantic relationships
     - Word2Vec uses this
  3. **Word embeddings:**
     - Dense vector representations
     - Similar words close together
     - Captures meaning
  4. **Contextual embeddings:**
     - BERT, GPT, etc.
     - Context-dependent representations
     - State-of-the-art

EXAM TRAP: n-grams, Skip-grams, and embeddings capture context!

Q76: TF-IDF vs Word Counts - Tutorial

QUESTION: What is the advantage of TF-IDF over simple word counts?

OPTIONS:

1. TF-IDF is faster to compute
2. TF-IDF assigns higher weights to rare but important words
3. TF-IDF counts all words equally
4. TF-IDF doesn't require a corpus

ANSWER: B

DETAILED TUTORIAL:

- **TF-IDF advantages:**
  1. **Weights rare words higher:**
     - Words appearing in few documents get high IDF
     - More discriminatory power
     - Better for distinguishing documents
  2. **Weights common words lower:**
     - Words in all documents get low IDF
     - No discriminatory power
     - Reduced noise
- **Comparison:**
  - Simple word count: treats all words equally
  - TF-IDF: distinguishes important from common words
  - Better feature representation

EXAM TRAP: TF-IDF = higher weight for rare words!

Q77: Document Similarity Use - Tutorial

QUESTION: How is document similarity measured in NLP?

OPTIONS:

1. By comparing document titles
2. By comparing document vectors using similarity metrics (cosine similarity)
3. By counting words in each document
4. By comparing document lengths

ANSWER: B

DETAILED TUTORIAL:

- **Similarity measurement:**
  1. **Document vectors:**
     - Create vectors for each document
     - Using BoW, TF-IDF, or embeddings
     - Vectors represent document content
  2. **Similarity metrics:**
     - Cosine similarity: $\cos(\theta) = \frac{v_1 \cdot v_2}{||v_1|| \times ||v_2||}$
     - Jaccard similarity
     - Euclidean distance
- **Applications:**
  - Duplicate detection
  - Plagiarism detection
  - Document clustering
  - Information retrieval

EXAM TRAP: Document similarity = comparing document vectors - often cosine similarity!

Q78: Cosine Similarity - Tutorial

QUESTION: What is cosine similarity used for in NLP?

OPTIONS:

1. To count word frequencies
2. To measure similarity between two document vectors
3. To remove stop words
4. To stem words

ANSWER: B

DETAILED TUTORIAL:

- **Cosine similarity formula:**
  - $\cos(\theta) = \frac{v_1 \cdot v_2}{||v_1|| \times ||v_2||}$
  - $v_1 \cdot v_2$: dot product
  - $||v||$: magnitude of vector
  - Range: -1 to 1
- **Advantages:**
  - Scale-invariant
  - Works well with high-dimensional vectors
  - Common in text analysis
  - Easy to interpret
- **Example:**
  - Document vectors are very similar $\rightarrow$ cosine equals 1
  - Document vectors are different $\rightarrow$ cosine equals 0
  - Opposite $\rightarrow$ cosine equals -1

EXAM TRAP: Cosine similarity = measures document vector similarity!

Q79: Stemming vs Lemmatization Tradeoff - Tutorial

QUESTION: What is the tradeoff between stemming and lemmatization?

OPTIONS:

1. No tradeoff - they are the same
2. Stemming is faster but less accurate; lemmatization is slower but more accurate
3. Stemming is more accurate
4. Lemmatization is faster

ANSWER: B

DETAILED TUTORIAL:

- **Stemming:**
  - **Advantages:**
    - Very fast
    - Simple rules
    - Low computational cost
  - **Disadvantages:**
    - Less accurate
    - May create non-words
    - Loses meaning
- **Lemmatization:**
  - **Advantages:**
    - More accurate
    - Always produces real words
    - Preserves meaning
  - **Disadvantages:**
    - Slower
    - More computationally expensive
    - Requires dictionary lookup
- **Choice depends on:**
  - Speed requirements
  - Accuracy requirements
  - Application context

EXAM TRAP: Stemming = fast but less accurate; Lemmatization = slow but more accurate!

Q80: NLP Challenges Summary - Tutorial

QUESTION: What are the main challenges in NLP?

OPTIONS:

1. Only computational speed
2. Ambiguity, context dependence, noise, and sarcasm/detecting tone
3. Only data size
4. Only language barriers

ANSWER: B

DETAILED TUTORIAL:

- **Key challenges:**
  1. **Ambiguity:**
     - Same words have different meanings
     - Polysemy and homonymy
     - Context determines meaning
  2. **Context dependence:**
     - Meaning depends on surrounding words
     - Need to capture relationships
     - Complex language structure
  3. **Noise:**
     - Spelling errors
     - Abbreviations
     - Colloquialisms
     - Social media language
  4. **Tone detection:**
     - Sarcasm hard to detect
     - Sentiment nuances
     - Emotional content

EXAM TRAP: NLP challenges include ambiguity, context, noise, and tone!

Q81: Sentiment Analysis Steps - Tutorial

QUESTION: What are the basic steps for sentiment analysis using a dictionary approach?

OPTIONS:

1. Only count positive words
2. Establish dictionary, pre-process text, replace with stems/lemmas, calculate proportions
3. Only count negative words
4. Use machine learning only

ANSWER: B

DETAILED TUTORIAL:

- **Sentiment analysis steps:**
  1. **Establish dictionary:**
     - Three lists: positive, negative, neutral
     - Loughran-McDonald for financial text
     - Pre-defined word lists
  2. **Pre-process text:**
     - Remove punctuation, symbols
     - Remove stop words
     - Convert to lowercase
  3. **Replace with stems/lemmas:**
     - Standardize word forms
     - Reduce vocabulary
     - Treat similar words equally
  4. **Calculate proportions:**
     - Proportion positive words
     - Proportion negative words
     - Proportion neutral words
     - Net sentiment = positive - negative

EXAM TRAP: Steps = dictionary, pre-process, stem, calculate proportions!

Q82: BoW Model Explanation - Tutorial

QUESTION: Explain how a bag of words model is used in NLP.

OPTIONS:

1. It preserves word order
2. It treats each term identically, counts frequencies, stores in vectors for analysis
3. It removes all words
4. It only works for numbers

ANSWER: B

DETAILED TUTORIAL:

- **BoW process:**
  1. **Treat terms identically:**
     - Ignore word order and position
     - Each word considered separately
     - No context information
  2. **Count frequencies:**
     - Count occurrences of each word
     - Binary (presence/absence) or count
     - Store in vector
  3. **Store in vectors:**
     - Each document = one vector
     - Vocabulary defines vector dimensions
     - Can be analyzed statistically
- **Applications:**
  - Dictionary-based sentiment analysis
  - Machine learning classification
  - Feature extraction for other models

EXAM TRAP: BoW = treats each word identically, counts, stores in vectors!

Q83: Naïve Bayes Assumptions - Tutorial

QUESTION: What are the two assumptions of Naïve Bayes for document grouping?

OPTIONS:

1. Words are ordered and have different weights
2. Each word is independent of others and each word has equal weighting
3. Words are dependent and have different weights
4. Words are random

ANSWER: B

DETAILED TUTORIAL:

- **Assumption 1: Independence:**
  - $P(w_1, w_2, \ldots, w_n|c) = P(w_1|c) \times P(w_2|c) \times \ldots \times P(w_n|c)$
  - Each word independent of all others
  - Conditional independence given class
- **Assumption 2: Equal weighting:**
  - Each word has equal information value
  - After removing stop words
  - No word is more important than others
- **Why these assumptions:**
  - Make the problem tractable
  - Work well in practice
  - Simplify calculations significantly

EXAM TRAP: Naïve Bayes = independence + equal weighting - two key assumptions!

Q84: Negation Problem and Solution - Tutorial

QUESTION: Why do negation words cause problems in document classification and how can they be handled?

OPTIONS:

1. They don't cause problems
2. Negations reverse meaning; use n-grams to pair negation with following words
3. Remove all negation words
4. Treat negation as positive

ANSWER: B

DETAILED TUTORIAL:

- **The negation problem:**
  - Example: "not adequate" vs "adequate"
  - BoW sees "adequate" in both
  - Meaning is opposite
  - Incorrect sentiment classification
- **n-gram solution:**
  - Use bi-grams or higher-order n-grams
  - Treat "not adequate" as one unit
  - "not sufficient" as one unit
  - "isn't good" as one unit
- **Alternative solutions:**
  - Negation detection algorithms
  - Dependency parsing
  - Identify scope of negation
  - Negation word + following word(s)

EXAM TRAP: Negation problem = solved with n-grams pairing negation + following word!

Q85: Zero TF-IDF Explanation - Tutorial

QUESTION: Why do some words have zero TF-IDF values?

OPTIONS:

1. Because they are misspelled
2. Because they appear in every document and have no discriminatory power
3. Because they are too long
4. Because they are not in the vocabulary

ANSWER: B

DETAILED TUTORIAL:

- **TF-IDF = 0 condition:**
  - IDF = ln(D / $\sum d_{i,j}$)
  - If word appears in all D documents: $\sum d_{i,j} = D$
  - IDF = ln(D/D) = ln(1) = 0
  - TF-IDF = TF $\times$ 0 = 0
- **Example:**
  - "this" appears in all 4 documents in Box 9.2
  - Zero discriminatory power
  - Cannot help distinguish documents
- **Implication:**
  - Common words carry no weight
  - Only rare words matter
  - Stop words often have zero TF-IDF

EXAM TRAP: Zero TF-IDF = word appears in all documents - no discriminatory power!

Q86: TF-IDF Variation by Document - Tutorial

QUESTION: Why does the same word have different TF-IDF scores in different documents?

OPTIONS:

1. Because IDF varies by document
2. Because TF varies by document (document length and frequency differences)
3. Because vocabulary changes
4. Because documents have different authors

ANSWER: B

DETAILED TUTORIAL:

- **TF variation:**
  - $TF = w_{i,j}/|v_j|$
  - $w_{i,j}$ = frequency of word in document
  - $|v_j|$ = total words in document
  - Different lengths $\Rightarrow$ different TFs
- **Example:**
  - Word appears 3 times in 100-word document: TF = 0.03
  - Word appears 3 times in 50-word document: TF = 0.06
  - Same word count, different TF
- **IDF is constant:**
  - IDF depends only on corpus
  - Same for all documents
  - TF varies $\Rightarrow$ TF-IDF varies

EXAM TRAP: TF varies by document (length, frequency) $\Rightarrow$ TF-IDF varies by document!

Q87: Appendix Problem Unknown Words - Tutorial

QUESTION: If new words appear in test documents not in the training sample, how should they be treated?

OPTIONS:

1. Add them to the vocabulary
2. Ignore them because they cannot help with classification
3. Remove the entire document
4. Use them as stop words

ANSWER: B

DETAILED TUTORIAL:

- **Unknown word handling:**
  - Words not in training vocabulary
  - No information about their distribution
  - Cannot calculate $P(w|c)$
  - Cannot contribute to classification
- **Why ignore them:**
  - They provide no discriminatory information
  - Would cause the model to fail
  - Better to exclude than to guess
- **Possible solution:**
  - Extend labeled sample to include new words
  - Use out-of-vocabulary handling
  - Build larger training vocabulary

EXAM TRAP: Unknown words = ignored - no information about them!

Q88: Smoothing Explanation - Tutorial

QUESTION: Why is smoothing required in Naïve Bayes applications and how does it work?

OPTIONS:

1. It's not required
2. To avoid zero probability problems by adding a positive integer to all counts
3. To remove stop words
4. To stem words

ANSWER: B

DETAILED TUTORIAL:

- **Why smoothing is needed:**
  - Zero probability problem
  - Word not in training class $\rightarrow$ P = 0
  - Product becomes zero
  - Unrealistic and draconian
- **How smoothing works:**
  - Add λ to all counts
  - $\lambda = 1$: Laplace smoothing (most common)
  - $P(w|c) = \frac{\text{count}(w,c)+\lambda}{\text{total words in c}+\lambda \times |V|}$
  - Ensures all probabilities > 0
- **Example:**
  - Before: $0/4 = 0$
  - After: $1/6 = 0.167$
  - Small positive probability

EXAM TRAP: Smoothing = add to counts to avoid zero probabilities!

Q89: Laplace Smoothing Impact - Tutorial

QUESTION: What happens to the probabilities under Laplace smoothing?

OPTIONS:

1. They become more extreme
2. They are pushed toward a uniform distribution
3. They become negative
4. They don't change

ANSWER: B

DETAILED TUTORIAL:

- **Effect of Laplace smoothing:**
  - Adds counts to all classes
  - All probabilities become positive
  - Shrinks toward uniform distribution
  - Reduces extreme values
- **How it works:**
  - Similar to regularization (Lasso/Ridge)
  - λ controls strength of smoothing
  - Larger λ = more smoothing
  - Smaller λ = less smoothing
- **Sample size effect:**
  - Small sample: large impact
  - Large sample: small impact
  - Effect diminishes with more data

EXAM TRAP: Laplace smoothing = shrinks toward uniform distribution!

Q90: Appendix Problem Classification - Tutorial

QUESTION: In the appendix problem, what is the most likely recommendation?

OPTIONS:

1. Buy (probability = 0.98)
2. Sell (probability = 0.02)
3. Hold
4. Cannot be determined

ANSWER: A

DETAILED TUTORIAL:

- **Given data:**
  - 5 buys, 5 sells (prior = 0.5 each)
  - New document: words 1, 2, 3 present; words 4, 5 absent
- **Conditional probabilities matrix:**
  - Word 1: P=4/5 (buy), 1/5 (sell)
  - Word 2: P=5/5 (buy), 2/5 (sell)
  - Word 3: P=2/5 (buy), 3/5 (sell)
  - Word 4: P=4/5 (buy, absent), 2/5 (sell, absent)
  - Word 5: P=3/5 (buy, absent), 1/5 (sell, absent)
- **Calculations:**
  - Buy likelihood = $0.8 \times 1.0 \times 0.4 \times 0.8 \times 0.6 \times 0.5 = 0.0768$
  - Sell likelihood = $0.2 \times 0.4 \times 0.6 \times 0.4 \times 0.2 \times 0.5 = 0.00192$
- **Result:** Buy (98% probability)

EXAM TRAP: Work through the full calculation - buy is much more likely!

Q91: NLP Pre-processing Steps - Tutorial

QUESTION: What are the main pre-processing steps in NLP?

OPTIONS:

1. Only tokenization
2. Tokenization, stop word removal, stemming/lemmatization, and feature extraction
3. Only stemming
4. Only stop word removal

ANSWER: B

DETAILED TUTORIAL:

- **Pre-processing steps:**
  1. **Tokenization:**
     - Split into sentences and words
     - Remove punctuation
     - Convert to lowercase
  2. **Stop word removal:**
     - Remove common words with no value
     - "the", "a", "and", etc.
  3. **Stemming/Lemmatization:**
     - Reduce words to root form
     - Standardize vocabulary
  4. **Feature extraction:**
     - Convert to numeric vectors
     - BoW, TF-IDF, embeddings

EXAM TRAP: Pre-processing = tokenization, stop words, stemming/lemmatization, feature extraction!

Q92: Tokenization Detail - Tutorial

QUESTION: What is the difference between sentence tokenization and word tokenization?

OPTIONS:

1. They are the same thing
2. Sentence tokenization splits at punctuation; word tokenization splits into words
3. Sentence tokenization removes words
4. Word tokenization removes sentences

ANSWER: B

DETAILED TUTORIAL:

- **Sentence tokenization:**
  - Splits document at periods (.)
  - Question marks (?)
  - Exclamation marks (!)
  - Creates sentence boundaries
- **Word tokenization:**
  - Splits sentences into words
  - Removes punctuation
  - Removes spacing and special symbols
  - Converts to lowercase
- **Example:**
  - Document: "Hello world. How are you?"
  - Sentence tokens: ["Hello world.", "How are you?"]
  - Word tokens: ["hello", "world", "how", "are", "you"]

EXAM TRAP: Sentence = splitting at punctuation; Word = splitting into words!

Q93: Stop Words Importance - Tutorial

QUESTION: Why do stop words vary across applications?

OPTIONS:

1. All stop words are the same
2. Words with no value in one context may be useful in another
3. Stop words are always removed
4. Stop words never matter

ANSWER: B

DETAILED TUTORIAL:

- **Context-dependent stop words:**
  - Some words may be useful in specific domains
  - Example: "bank" is common but important in financial text
  - "stock" is common but important in market analysis
- **Why customization matters:**
  - Standard stop word lists may remove important words
  - Domain-specific vocabulary is important
  - Need to consider the application
- **Examples:**
  - Medical NLP: preserve all medical terms
  - Financial NLP: preserve financial terms
  - Legal NLP: preserve legal terms

EXAM TRAP: Stop words are context-dependent - not one-size-fits-all!

Q94: DTM Example - Tutorial

QUESTION: In Figure 9.1, what does the Document Term Matrix show?

OPTIONS:

1. Document titles and authors
2. Rows as documents, columns as words, entries as word counts
3. Document dates
4. Document sources

ANSWER: B

DETAILED TUTORIAL:

- **DTM structure:**
  - **Rows:** Each row = one document
  - **Columns:** Each column = one word from vocabulary
  - **Entries:** Word counts or frequencies
  - Dimensions: $|W| \times D$
- **Example from Figure 9.1:**
  - Vocabulary: 15 words (high, low, buy, sell, etc.)
  - Documents: d1 and d2
  - d1: high appears twice, stock appears once, etc.
  - d2: bond appears once, volatility appears once, etc.
- **Key features:**
  - Same number of columns for each document
  - Enables quantitative analysis
  - Sparse (many zeros)

EXAM TRAP: DTM = documents $\times$ words matrix with counts!

Q95: Feature Extraction Purpose - Tutorial

QUESTION: Why is feature extraction (text representation) important in NLP?

OPTIONS:

1. To make text longer
2. To convert text into numeric vectors for machine learning analysis
3. To remove words
4. To make text more readable

ANSWER: B

DETAILED TUTORIAL:

- **Feature extraction purpose:**
  - Machine learning models need numeric inputs
  - Text must be converted to numbers
  - Enables statistical analysis
  - Foundation for all ML-based NLP
- **Common methods:**
  1. **Bag of Words:**
     - Simple word counts
     - Binary or count
  2. **TF-IDF:**
     - Weighted word importance
     - Rare words get higher weight
  3. **Word embeddings:**
     - Dense vector representations
     - Captures semantic meaning

EXAM TRAP: Feature extraction = text $\rightarrow$ numeric vectors!

Q96: Document Term Matrix Dimensions - Tutorial

QUESTION: What are the dimensions of a Document Term Matrix?

OPTIONS:

1. $D \times W$ (documents $\times$ words)
2. $W \times D$ (words $\times$ documents)
3. $D \times D$ (documents $\times$ documents)
4. $W \times W$ (words $\times$ words)

ANSWER: A

DETAILED TUTORIAL:

- **DTM dimensions:**
  - Rows: D = number of documents
  - Columns: W = number of words in vocabulary
  - Dimensions: D $\times$ W
  - Entry: word count or frequency
- **Example:**
  - 100 documents
  - 10,000 word vocabulary
  - Matrix size: 100 $\times$ 10,000
  - 1,000,000 entries
- **Implications:**
  - Very large matrices
  - Sparse (mostly zeros)
  - Need efficient storage

EXAM TRAP: DTM dimensions = documents $\times$ words!

Q97: BoW Vector Example - Tutorial

QUESTION: In the BoW example, why does d1 have a 2 for "high" in the count vector?

OPTIONS:

1. Because "high" is the most important word
2. Because "high" appears twice in the document
3. Because "high" is a stop word
4. Because "high" is the first word

ANSWER: B

DETAILED TUTORIAL:

- **Example text:**
  - d1: "... reaching a high of 1550 ... with a high of 1560"
  - "high" appears twice
  - Count vector records 2
- **Count vs Binary:**
  - **Count:** 2 (frequency)
  - **Binary:** 1 (presence)
  - Count gives more information
- **Other words:**
  - "stock" appears once $\rightarrow$ 1
  - "profit" appears once $\rightarrow$ 1
  - "rise" appears once $\rightarrow$ 1
  - "fall" appears once $\rightarrow$ 1

EXAM TRAP: Count BoW = actual frequency - not just presence!

Q98: Bag of n-grams - Tutorial

QUESTION: What is the Bag of n-grams (BoN) approach?

OPTIONS:

1. A method to count single words
2. Applying BoW to n-grams (n contiguous words as units)
3. A method to remove n-grams
4. A method to count only bigrams

ANSWER: B

DETAILED TUTORIAL:

- **BoN definition:**
  - Extends BoW to n-grams
  - n contiguous words = one unit
  - Example: "credit spreads" as one unit
  - Handles phrases and negations
- **Example:**
  - Document: "The bank is not profitable"
  - Bi-grams: ["The bank", "bank is", "is not", "not profitable"]
  - "not profitable" as one unit
- **Advantages:**
  - Captures phrase meaning
  - Handles negations
  - Better context information

EXAM TRAP: BoN = BoW applied to n-grams - captures phrases!

Q99: Cosine Similarity Application - Tutorial

QUESTION: In document similarity, what does a cosine similarity close to 1 indicate?

OPTIONS:

1. Documents are very different
2. Documents are very similar (vectors point in same direction)
3. Documents are opposite
4. Documents are unrelated

ANSWER: B

DETAILED TUTORIAL:

- **Cosine similarity interpretation:**
  - **cos equals 1:** Very similar documents
  - **cos equals 0:** Unrelated documents
  - **cos equals -1:** Opposite documents
  - Range: [-1, 1]
- **Why it works:**
  - Vectors point in same direction
  - Similar word distributions
  - Angle between vectors is small
  - Magnitude doesn't matter (scale-invariant)
- **Applications:**
  - Finding duplicate documents
  - Information retrieval
  - Document clustering
  - Plagiarism detection

EXAM TRAP: cos equals 1 = very similar documents!

Q100: NLP Chapter Summary - Tutorial

QUESTION: What are the key takeaways from the NLP chapter?

OPTIONS:

1. Only BoW is used in NLP
2. NLP involves pre-processing, feature extraction, dictionary or ML approaches, and careful evaluation
3. Only machine learning is used
4. NLP doesn't require pre-processing

ANSWER: B

DETAILED TUTORIAL:

- **Key takeaways:**
  1. **Pre-processing:**
     - Tokenization
     - Stop word removal
     - Stemming/lemmatization
     - Feature extraction
  2. **Feature extraction:**
     - Bag of Words (BoW)
     - n-grams
     - TF-IDF weighting
  3. **Approaches:**
     - Dictionary-based (heuristic)
     - Machine learning (Naïve Bayes, SVM, neural networks)
     - Word embeddings
  4. **Evaluation:**
     - Confusion matrix metrics
     - Speed and data requirements
     - Context-aware techniques

EXAM TRAP: NLP = pre-processing + feature extraction + modeling + evaluation!

Summary: Complete NLP Review

Pre-processing:

- Tokenization: Sentences $\rightarrow$ words
- Stop Words: Remove common words with no value
- Stemming: Cut prefixes/suffixes (fast, less accurate)
- Lemmatization: Dictionary form (slow, more accurate)

Feature Extraction:

- BoW: Binary (presence) or Count (frequency)
- n-grams: Contiguous word sequences
- TF-IDF: $TF \times IDF$ - weights rare words higher
- Embeddings: Dense vectors capturing meaning (Word2Vec)

Classification:

- Dictionary: Pre-defined word lists (sentiment analysis)
- Naïve Bayes: Simple, fast, generative classifier
- Smoothing: Laplace smoothing to avoid zero probabilities

Evaluation:

- Confusion Matrix: TP, TN, FP, FN
- Metrics: Accuracy, Precision, Recall, F1
- Challenges: Ambiguity, context, negation, sarcasm

Exam Success: Understand the process from raw text to classification and evaluation!